Changkai Chen

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Research profile

Astronomy Ph.D. student studying black-hole perturbation theory, gravitational radiation, and quasinormal spectra. Combines analytic methods and high-precision numerical calculations to develop and benchmark tools for Kerr-orbit fluxes, black-hole spectroscopy, and strong-field gravity.

Research interests

Black-hole perturbation theory; gravitational-wave source modelling; Kerr-orbit radiation; quasinormal modes and scattering; numerical methods for relativistic astrophysics.

Education

Beijing Normal University, Beijing, China 2025–present

Ph.D. in Astronomy, expected June 2027. Supervisor: Prof. Zhoujian Cao.

Hunan Normal University, Changsha, China 2021–2024

M.Sc. in Astronomy. Supervisor: Prof. Jiliang Jing.

Thesis: Radiation fluxes of perturbation fields in type-D black holes.

China Three Gorges University, Yichang, China 2016–2020

B.Sc. in Physics. Supervisor: Prof. Xiaohua Zhang.

Research appointment

Research Assistant, Hunan Normal University, Changsha, China 2024–2025

Research contributions

Efficient gravitational-wave fluxes for generic Kerr orbits

Developed a unified confluent-Heun treatment of the angular and radial Teukolsky equations, with hybrid analytic continuation for connection coefficients and adaptive bipower mapping quadrature for oscillatory source integrals. In the published 168-mode benchmark, obtained relative errors of order 10^{-11} in double precision and achieved typical speedups of 3–13 over GeneralizedSasakiNakamura.jl and PYBHPT; selected highly oscillatory modes achieved speedups up to 60. [1]

Black-hole quasinormal spectra and scattering

Developed an analytic-continuation method that computes modes crossing or lying on the negative imaginary axis. Presented Schwarzschild and Kerr spectra over $0 \le n \le 41$ and $2 \le \ell \le 16$, validated through scattering amplitudes with errors below 10^{-10}; resolved the apparent small-spin discontinuity and clarified the relation to algebraically special and unconventional modes. [2]

Strong-field radiation from eccentric and inclined orbits

Extended the Heun-function approach from circular Schwarzschild orbits to generic orbits around near-extreme Kerr black holes, including benchmark cases with eccentricity $e = 0.9$ and dimensionless spin $a/M = 0.999$. Computed asymptotic fluxes and compared accuracy with MST-based numerical integration. Earlier work treated gravitational, electromagnetic, and scalar radiation. [3,4]

Total-transmission modes and exceptional points

Constructed four globally continuous Kerr total-transmission-mode families by separating mirror-related sectors in an earlier classification, without adding symmetry-unrelated roots. Studied spectra up to $l = 32$, exceptional-point coalescence, and a representative TTM–QNM proximity of order 10^{-8}. Preprint. [7]

High-order numerical methods for multidimensional PDEs

Combined sixth-order compact spatial differences with precise integration and Strang splitting for multidimensional parabolic equations. Extended this approach to Burgers systems using a modified Hopf–Cole transformation; tested accuracy against analytical solutions in one, two, and three dimensions. Example codes were released with the Burgers paper. [5,6]

Peer-reviewed publications

Six first-author articles. Numbers correspond to research descriptions.

[1] Changkai Chen, Zhoujian Cao, and Jiliang Jing. Efficient and stable computation of gravitational-wave fluxes from generic Kerr orbits via a unified Heun-function framework. Physical Review D 114, 063002 (2026). DOI.

[2] Changkai Chen, Jiliang Jing, Zhoujian Cao, and Mengjie Wang. Complete quasinormal modes of type-D black holes. Physical Review D 112, 103036 (2025). DOI.

[3] Changkai Chen and Jiliang Jing. Gravitational wave fluxes on generic orbits in near-extreme Kerr spacetime: Higher spin and large eccentricity. Science China Physics, Mechanics & Astronomy 67, 110411 (2024). DOI.

[4] Changkai Chen and Jiliang Jing. Radiation fluxes of gravitational, electromagnetic, and scalar perturbations in type-D black holes: an exact approach. Journal of Cosmology and Astroparticle Physics 11 (2023) 070. DOI.

[5] Changkai Chen, Xiaohua Zhang, and Zhang Liu. A high-order compact finite difference scheme and precise integration method based on modified Hopf-Cole transformation for numerical simulation of n-dimensional Burgers' system. Applied Mathematics and Computation 372, 125009 (2020). DOI.

[6] Changkai Chen, Xiaohua Zhang, Zhang Liu, and Yage Zhang. A new high-order compact finite difference scheme based on precise integration method for the numerical simulation of parabolic equations. Advances in Difference Equations 2020, 15 (2020). DOI.

Preprint

[7] Changkai Chen, Xiaohua Zhang, Zhoujian Cao, Jiliang Jing, and Sheng Long. Complete total-transmission modes of Kerr black holes. arXiv:2609.39609 [gr-qc] (2026), preprint. arXiv.

Computational methods and tools

Black-hole perturbation and scattering calculations; analytic continuation; oscillatory quadrature; compact finite differences; precise integration; Strang splitting.

Languages and tools: Mathematica, MATLAB, Python, C/C++, Maple, and LaTeX.

Selected presentations and training

New Exact Approach of Gravitational Radiation, presentation, Annual Meeting on Gravitation and Relativistic Astrophysics, Chinese Physical Society, Chongqing University, April 2023.

High-Precision Numerical Simulation of Shock Waves, presentation, 2nd Advanced Workshop on Numerical Methods for Differential Equations, Tianyuan Mathematical Center in Central China, March 2021.

Gravitational-wave summer schools: TianQin Center, Sun Yat-sen University (2023); Taiji Center, International Centre for Theoretical Physics Asia-Pacific (2022).

Selected awards

Selected participant, China Association for Science and Technology (CAST) Young Science and Technology Talent Development Program (Doctoral Student Special Program), 2026.

First Prize Scholarship and Outstanding Postgraduate Student, Hunan Normal University (2021–2023).

National First Prize, China Postgraduate Mathematical Contest in Modeling (2021).

National First Prize, CSEE Cup Mathematical Modeling Competition (2018).

Outstanding Graduate, China Three Gorges University (2020).